MIL-C-24643/53B
3 October 1986
SUPERSEDING
MIL-C-0024643/53A(SH)
8 November 1985
MIL-C-24643/53
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 3000 VOLTS, TYPES LSSRW, LSDRW, AND LSTRW (INCLUDING VARIATIONS LSSRWA, LSDRWA, AND LSTRWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in the following variations: LSSRW, LSDRW, and LSTRW (unarmored) and LSSRWA, LSDRWA, and LSTRWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated. AWG size 14, class B stranding.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation, colored black. Nominal thickness 0.080 inch. Standard identification code applied by method 1, on two-conductor and three-conductor cables only.
Fourth - The required two or three conductors cabled with a lay not greater than 24 times the pitch diameter. Fillers shall be employed, as necessary, to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap for two and three conductor cables only.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for types LSSRW, LSDRW and LSTRW

Sixth - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch. Cable surface marking required.

NOTE: Manufacturer's marker tape may be omitted from single conductor cable.

Construction variant for types LSSRWA, LSDRWA, and LSTRWA

(B) Sixth - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch. Cable surface marking not required.
Seventh - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/53	Cable type	Number of conductors	Overall diameter maximum (inch)	Conductor resistance at 25°C max 1000 feet (ohms)	Cold bending mandrel maximum diameter (inches)
-01UN	LSSRW	1	0.400	2.69	3
-02UN	LSDRW	2	.670	2.75	5
-03UN	LSTRW	3	.710	2.75	7
-04UN	LSSRWA	1	.450	2.69	3
-05UN	LSDRWA	2	.720	2.75	5
-06UN	LSTRWA	3	.760	2.75	7

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table I)
Voltage withstand - volts, root mean square, minimum....	8000
Insulation resistance - megohms/1000 feet, minimum.....	9500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for water leakage.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - (See table I for mandrel diameter)	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductor) (types LSDRW and LSTRW) cyles, minimum	25
Permanence of printing (jacket), types LSSRW, LSDRW and LSTRW - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to material, construction and coverage, types LSSRWA, LSDRWA and LSTRWA only	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Types	Feet (nominal)
LSSRW	2500
LSDRW and LSTRW	2000

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-53)

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